- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**9** A particle is moving in a straight line. The acceleration $a\,\text{m s}^{-2}$ of the particle at time $t$ s is given by $a = 0.8t + 0.5$. The initial velocity of the particle is $3\,\text{m s}^{-1}$ in the positive $x$-direction.

Determine whether the particle is ever stationary. **[6]**

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